NOISEMAP · RISK REGISTER AND HAZARD ANALYSIS

NoiseMap risk register and hazard analysis

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.1	2026-11-02	Amish Chadha	First issue (kit 1.8 pipeline), with the first-pass design FMEA at stage P3

NoiseMap risk register and hazard analysis

Top risks: measurement quality that paper cannot settle (R-2) and a privacy promise that rests on firmware, not physics (R-5); the top hazards are work at height beside traffic (H1) and a falling arm or head (H2). The FMEA's highest numbers are the hydrophobic membrane (F6, RPN 120), the level firmware (F10, 105), the V-block bands (F1, 96), and the lanyard and privacy firmware (F4 and F11, 90 each). Method: STANDARDS section 23. Likelihood, impact, severity and occurrence are scored 1 (low) to 5 (high); detection 1 (certain to be caught) to 5 (not detectable at TRL 3). All scores are the author's proposals, "Proposed, awaiting Amish".

Project risk register

Table 1. Project risks.

#	Risk	Cause	Effect	Likelihood	Impact	Owner (role)	Response
R-1	Co-design partner not yet agreed (NSM-PRB-001 open items)	No partner has confirmed the user needs	UR priorities and targets stay proposals	3	4	Amish	Approach a city noise team or university acoustics group (decided 2026-10-02); revise URS and PRS from their findings
R-2	Measurement quality (PR-003 to PR-005) misses its targets	Membrane, windscreen and head diffraction unknown; facade bias	Accuracy claim cannot be made	4	4	Design lead	Reference meter at TRL 4 (on hold); keep every claim 'indicative'
R-3	Mass margin of 0.05 kg is used up	A heavier windscreen or an anemometer is added	PR-010 not met	3	3	Design lead	Mass check on any added part (NSM-DDR-003 A2)
R-4	ICS-43434 supply (end of life)	Maker listing	Microphone has to change	4	2	Design lead	Dual-footprint adapter board takes the IM72D128 (D3)
R-5	Privacy claim is doubted	Firmware could in principle send audio; the cable could carry a speech codec	Loss of trust, no deployment	3	4	Amish	Open firmware, read-out protection, build hash; never claim privacy by physics (D2)
R-6	Regulatory unknowns (radio, GDPR, battery)	Items marked to assess in NSM-REG-001	Delay before any deployment	3	3	Regulatory consultant	Consult before any pilot; no product claim
R-7	Options A and B within 10 %: choice delayed	Score gap 4.7 %	Wait for Amish	3	2	Amish	Decision request in NSM-PLN-001
R-8	TRL cap	Portfolio capped at TRL 3	No testing to settle the at-risk items	5	2	Amish	Plan only; P4 on hold
R-9	LoRaWAN airtime margin at SF12	At SF12 the node uses 29.97 s of the 30 s a day it may transmit, leaving no room for time requests, joins or retries	Records lost or delayed at the cell edge	3	3	Design lead	Interval rule (NSM-DDR-002 O4); SF7 to SF9 expected in town; PR-002 and PR-012 stated for SF7 to SF9 only (NSM-DEC-001 decision 3)

Preliminary hazard analysis

Table 2. Hazards. Option A unless stated; the mains hazard H11 belongs to option C only.

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H1	Work at height beside traffic	Fitting or servicing a node 4 m up a street pole	Fall; person struck by vehicle	5	3	Trained crew, owner's permission, fall protection, traffic management (NSM-PRC-001)	Medium
H2	Falling part	Clamp slips, tube pulls out, wire rope fails	Head injury to a passer-by from 4 m	5	2	Two band clamps, cross bolts with nyloc nuts, lanyard (PR-017), check after the first storm	Low to medium
H3	Wind load on the pole	35 m/s gust on 0.1 m2 of panel, box and windscreen	Pole overload (about 96 N at 3.5 to 4 m)	4	2	Pole owner confirms the pole can take it; no more area without a check	Low
H4	Lithium cell fault	Damaged, swollen or cold-charged cell	Heat, fire	4	2	Fuse, NTC, cold-charge lockout, no swollen cell; FieldNode core item	Low
H5	Overhead lines and pole wiring	Reaching or opening a pole hatch	Electric shock	5	2	Keep clear of overhead lines; never open a pole's electrical hatch; no electrical connection to the pole	Low
H6	Hearing damage	Calibrator at 94 dB held near the ear	Temporary threshold shift	2	2	Hearing protection if the calibrator sounds for more than a short check; 2 minutes against 60 allowed (STM S19)	Low
H7	Sharp edges	Sawn aluminium bar, cut sheet, wire rope ends	Cuts	2	3	File and break edges (modelled 0.5 mm); gloves during build	Low
H8	Privacy harm	Audio stored or sent, or believed to be	Loss of trust, complaints, legal exposure	4	2	Levels only by design; open firmware with a build hash; notice plate (PR-016); data protection review (NSM-REG-001 G6)	Medium until reviewed
H9	Readings misused as legal evidence	Indicative data presented as certified	Wrong enforcement decision	3	3	Marked as indicative in every document; certified co-located meter needed for enforcement; out of scope	Low
H10	Birds and the bird spike	Spike on the windscreen	Soft wording: harm to a bird	2	2	Blunt-tipped 3 mm rod, placed off the microphone axis; effect not claimed	Low

#	Hazard	Cause	Harm	Severity	Likelihood	Control	Residual
H11	Option C only: mains voltage in the puck	Plugging into a live luminaire receptacle	Electric shock	5	3	Qualified electrician; isolated converter; not part of option A	Medium (not carried forward in A)

> **Safety:** installing on a street pole is work at height next to traffic, and the node carries a lithium cell and a falling-part risk. See the safety section of NSM-PRC-001.

Design FMEA (first pass, P3)

Table 3. Failure modes of option A. RPN = severity x occurrence x detection. Scores are first-pass paper estimates; none is measured.

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F1	V-blocks and bands	Hold the core to the pole	Band slips or breaks	Core rotates or falls	Preload loss, corrosion, vibration	8	3	4	96	Torque to a stated value; check after the first storm (NSM-PRC-001 safety note); keep preload 1,000 N (NSM-CAL-001 H6)
F2	Arm saddle and bands	Hold the arm to the pole	Saddle fatigue crack or band slip	Arm and head fall 4 m	Wind cycling, sheet bend cracks	9	2	4	72	Lanyard (PR-017); 2.5 mm 5052-H32 sheet bent across the grain's safe radius (to confirm with the shop)
F3	Arm tube and cross bolts	Carry the head	Tube pulls out of flange or socket	Head hangs on lanyard	Loose cross bolt	8	2	3	48	Nyloc nuts and threadlocker (BLD Step 5 and 8); lanyard
F4	Safety lanyard	Retain the arm	Wire rope or ferrule fails	Arm falls	Wrong ferrule or nick in the wire	9	2	5	90	Four ferrules and two thimbles by the plan; inspect at each foam change
F5	Head housing	Keep the electronics dry and fixed	UV cracking or warping of the printed ASA	Water in the head	UV, heat cycling, print layer weakness	6	3	4	72	Print orientation and 3 mm wall (NSM-DDR-003); inspect yearly
F6	Hydrophobic membrane	Keep water out of the port while passing sound	Clogs or tears	Water at the microphone; level error	Dirt, bird contact, age	6	4	5	120	Replace with the foam; port protected by the windscreen
F7	Foam windscreen	Reduce wind noise	Degrades or soaks	Wind noise, level bias	UV, rain	4	5	3	60	Yearly change (PR-014)
F8	Cable and gland	Carry levels and 3.3 V	Chafing or gland leak	Loss of data; water entry	Rubbing on the arm, wrong cable diameter	6	3	4	72	Cable ties, gland sized to 4 to 8 mm cable; route in BLD Step 12
F9	LiFePO4 cell and charger	Store solar energy	Charged below 0 degrees C	Plating, capacity loss, in rare cases a safety event	Charger lockout fails	7	2	4	56	FieldNode cold-charge lockout and fuse; do not fit a swollen cell

#	Item	Function	Failure mode	Effect	Cause	S	O	D	RPN	Action
F10	Level processor firmware	Compute A and C levels	Wrong gain or filter	Biased levels reported as true	Coefficient or scaling error	7	3	5	105	Open firmware; field check with a 94 dB calibrator at install (PR-009); site correction
F11	Firmware privacy	Send levels only	A code path sends samples	Loss of trust, data protection exposure	Added debug feature	9	2	5	90	Read-out protection, published build hash, review of every release (PR-001)
F12	MEMS microphone	Convert sound to a signal	Clips above its overload point (119 dB SPL for the IM72D128, 113 for the ICS-43434) or sits at its noise floor	Wrong peak or night level	Overload; self-noise at 35 dBA	5	4	3	60	Use the IM72D128 where quiet levels matter (PR-004); flag clipped seconds
F13	Solar panel	Charge the cell	Soiling or shading	Less harvest	Dust, bird droppings	3	4	3	36	Panel doubles as a hood; 20 times margin on energy (NSM-CAL-001 E1)
F14	LoRaWAN link	Send the record	No gateway or join fails	Records stored in flash, then late	Poor coverage, antenna	3	4	2	24	Flash store and forward; gateway survey before install
F15	Clock	Time-stamp records	Drift over weeks	Records misaligned with events	Crystal drift	3	3	3	27	LoRaWAN network time resync (PR-006)